

Exp No: 14:

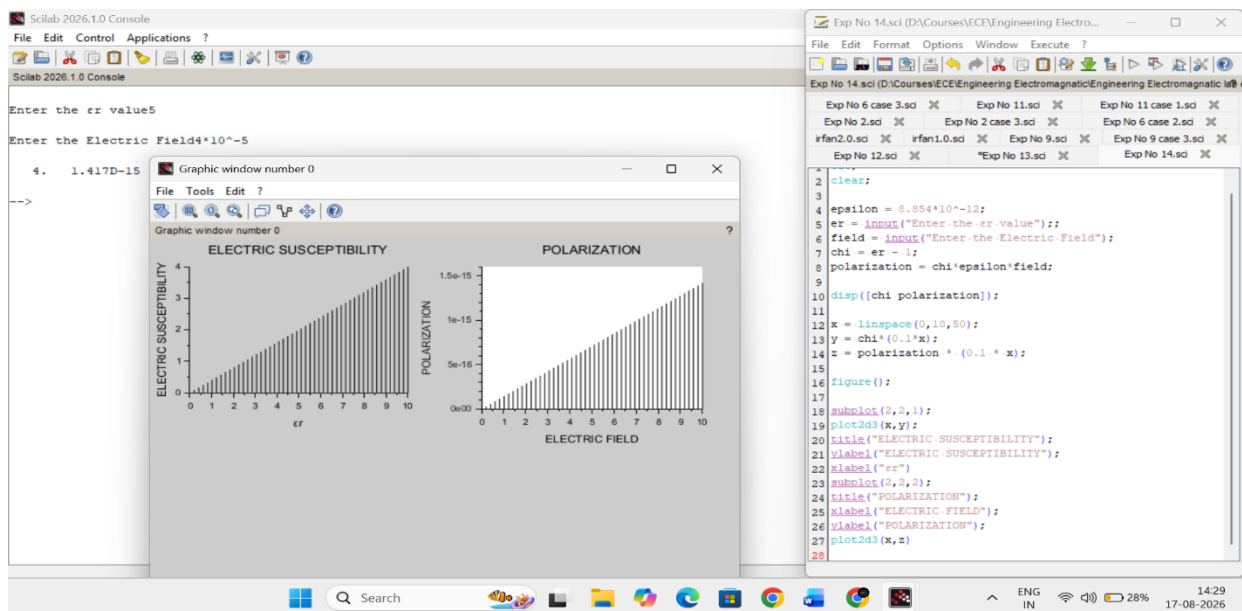
Determination and plotting of Polarization in different Dielectric Medium with their relative permittivity and to find electric susceptibility.

Date:

Aim: To determine and plot polarization in different dielectric medium with their relative permittivity and to find electric susceptibility.

Software required:

- SCILAB version 6.0.1



Exp No: 14

Finding electric susceptibility

Polarization

$$\epsilon_r = 5$$
$$E = 4 \times 10^{-5}$$

Electric susceptibility

$$\chi_e = \epsilon_r - 1$$
$$= 5 - 1$$

$$\boxed{\chi_e = 4}$$

d.p.v
Polarization

$$P = \chi_e \epsilon_0 E$$
$$= (4) (8.854 \times 10^{-12}) (4 \times 10^{-5})$$

$$= 4 \times 8.854 \times 10^{-12} \times 4 \times 10^{-5}$$

$$= 3.5416 \times 10^{-11} (4 \times 10^{-5})$$

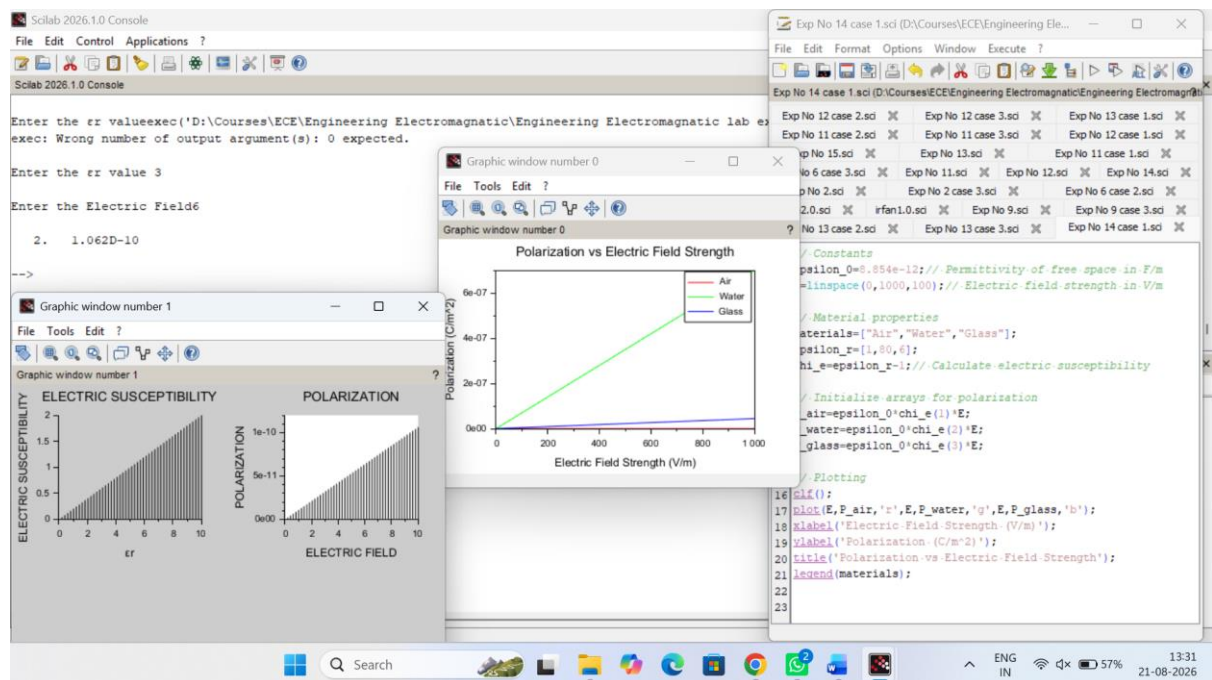
$$P = 1.4166 \times 10^{-15}$$

$$\boxed{P = 1.417 \times 10^{-15}}$$

Result:

Thus, the determination and plotting of polarization in different dielectric medium with their relative permittivity and to find electric susceptibility is executed.

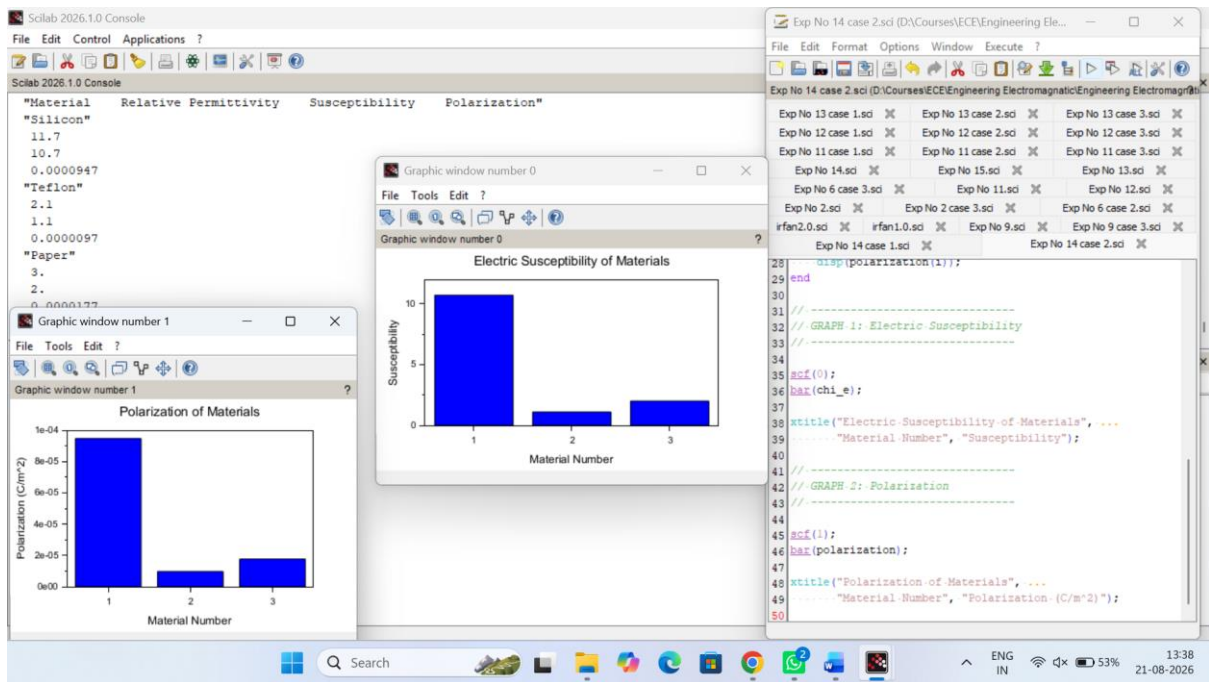
Case 1: To determine and plot the polarization and electric susceptibility of electromagnetic waves in three dielectric media: Air, Water, and Glass, using their relative permittivity.



Result:

Thus, the determination and plotting of polarization in different dielectric medium with their relative permittivity and to find electric susceptibility is executed.

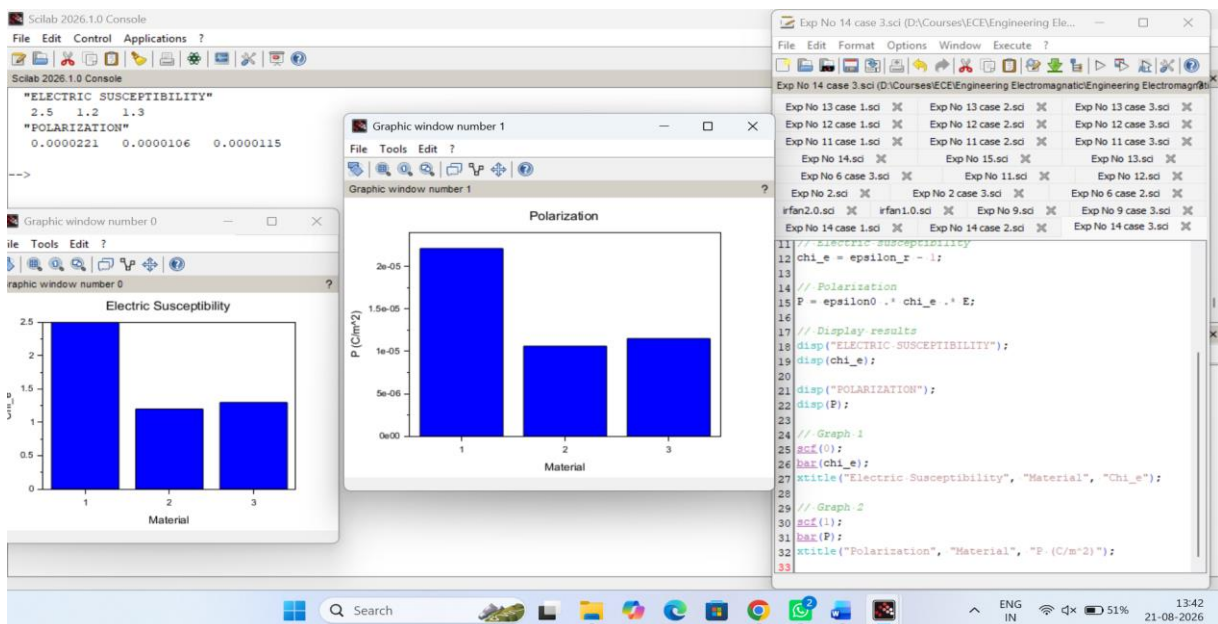
Case 2: To calculate and plot the polarization and electric susceptibility in Silicon, Teflon, and Paper, and understand how their relative permittivity affects the polarization.



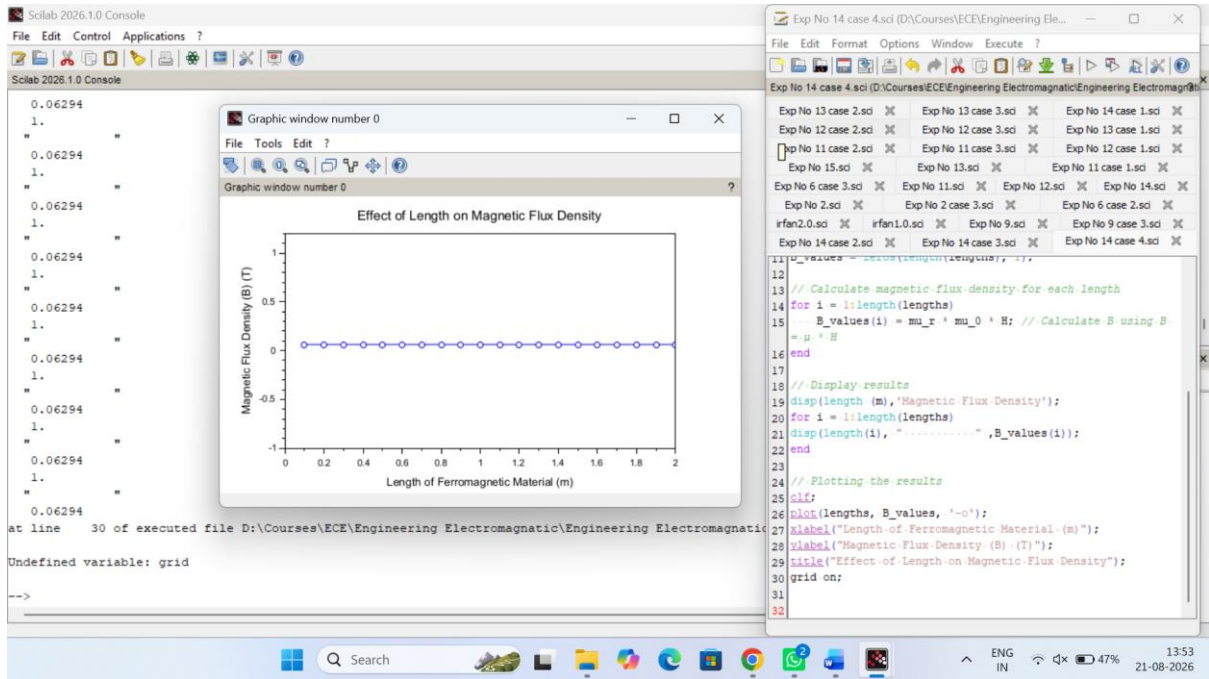
Result:

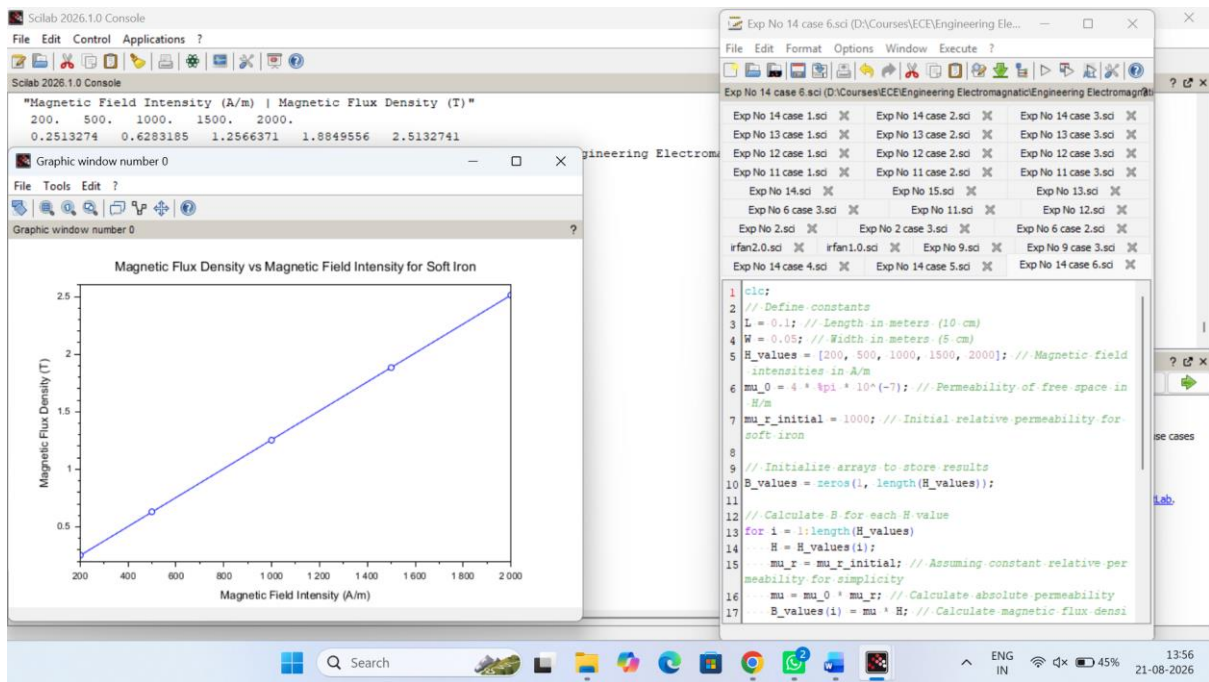
Thus the determination and plotting of polarization in different dielectric medium with their relative permittivity and to find electric susceptibility is executed.

Case 3: To explore how different dielectric materials (Rubber, Oil, and Plastic) affect the polarization and electric susceptibility, and to compare their relative permittivity's.



Case 1: Examine the effect of length of a ferromagnetic material on the magnetic flux density (B) for the following conditions





Result:

Thus the program was verified for program for Magnetic flux density in ferromagnetic materials using SCILAB was successfully.